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Nonlinear Dynamics in Structural Systems

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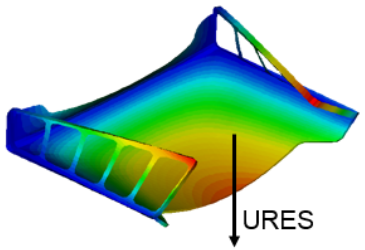
CE 534 – Structural Dynamics

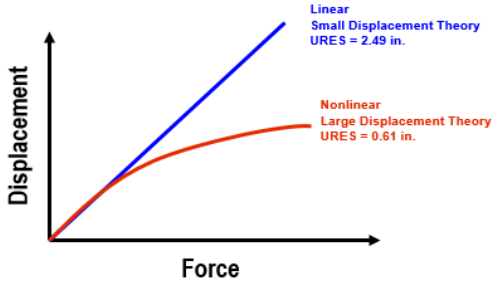
Overview of Nonlinear Dynamics

What Does "Nonlinear" Mean?

A system is nonlinear when its response is not proportional to the applied load.

- **Linear system:** Double the load → double the response
- **Nonlinear system:** Double the load → response may triple, plateau, or behave unpredictably



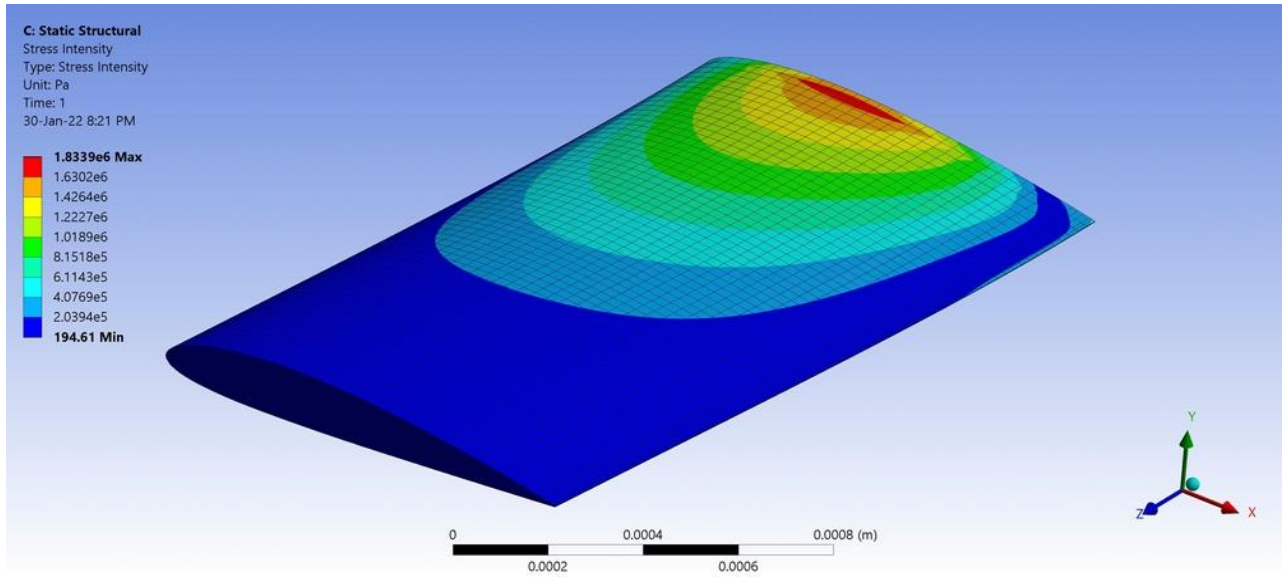


Nonlinear dynamics refers to systems whose behavior over time cannot be described by a simple linear relationship between inputs and outputs

Why Do Structures Behave Nonlinearly?

Nonlinear behavior appears when loads become large enough to activate effects such as:

- Material nonlinearity:
  Concrete cracking, steel yielding, joint slip
- Geometric nonlinearity:
  Large deflections change the structure's stiffness
- Boundary condition nonlinearity:
  Supports or connections that shift, slip, or open/close



Case Study: Jump Around at Camp Randall

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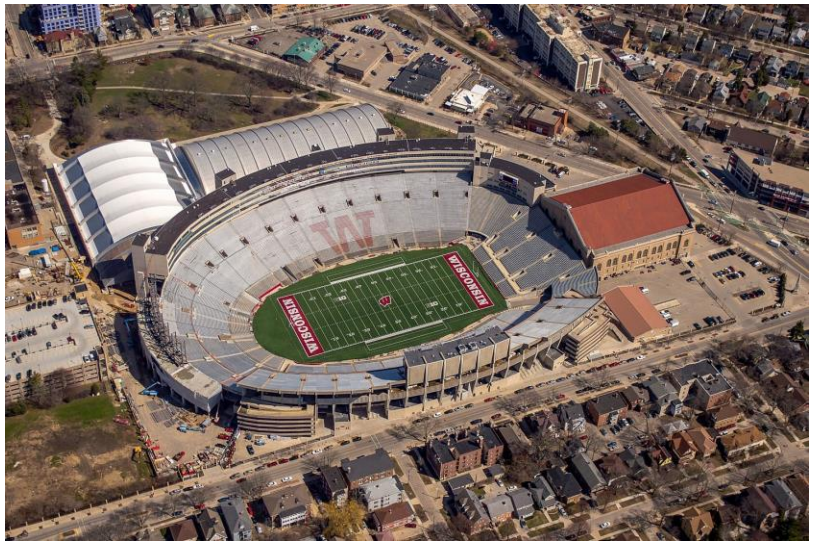
- Camp Randall Stadium is one of the oldest stadiums in college football. Its initial construction was completed in 1917
- The upper deck was built with reinforced concrete in 1966.
- **Structure**: It features 23 large frames, and the seating is supported by a cantilevered design that extends from a single column on each frame.
- **Function**: This cantilevered design allows the seating area to extend over the lower decks.
- The concrete bowl and precast risers/rakers form a fairly stiff, heavy, continuous structure. Under dynamic loading, this mass and stiffness provides inertia and damping which is good for resisting vibrations.
- The "Jump Around" tradition started in 1998
- Over 80,000 fans jump in unison to the House of Pain song
- Causes noticeable stadium vibrations in the upper deck



Why nonlinear dynamics is important:

Crowds can act as synchronized dynamic loads, which amplifies structural motion.

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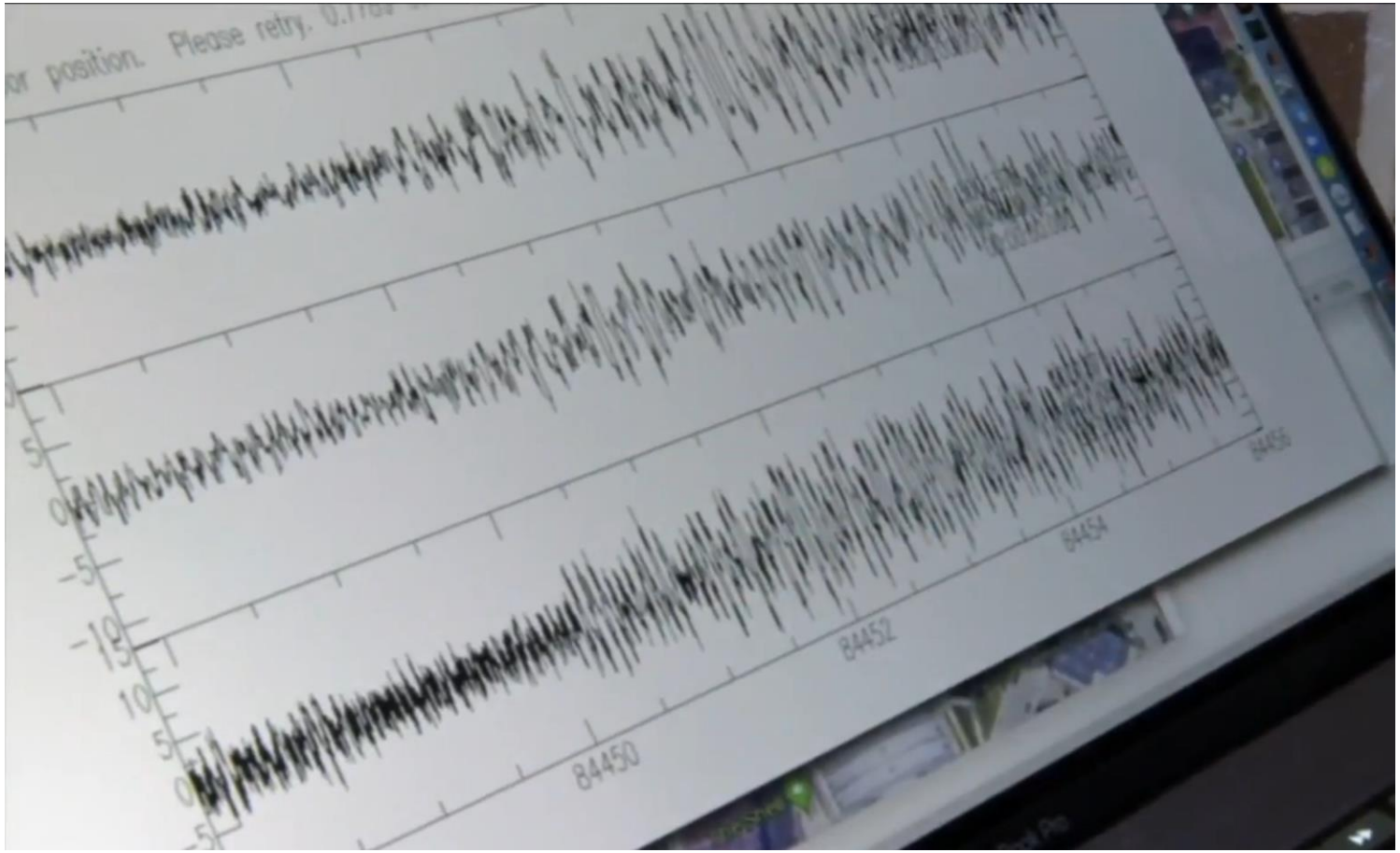
A geophysicist at the University of Wisconsin measured the vibration frequencies of the stadium periodically and found that the stadium vibrations were shaking the ground at a frequency that the stadium itself is vibrating whenever the fans jump around at the end of the 3rd quarter



Structural Design of Upper Deck

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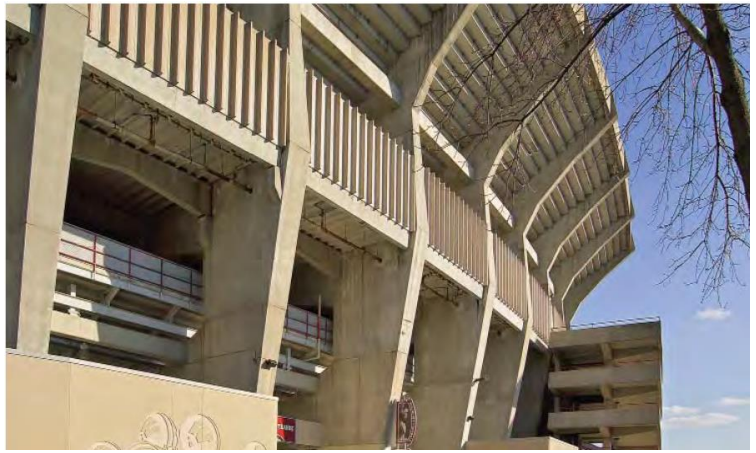
- The upper deck was modeled as a 3D space frame with concrete slabs represented using plate elements. The final dynamic model contained 292 degrees of freedom



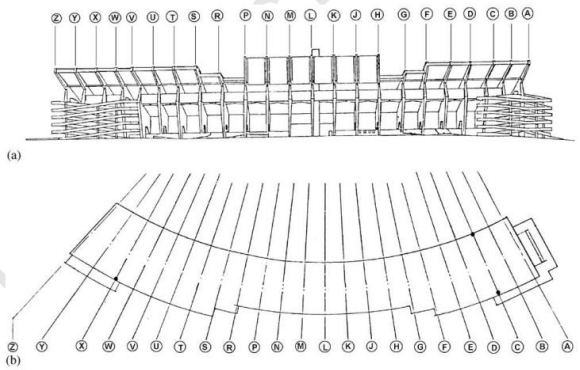


Fig. 1. (a) View toward the back of the University of Wisconsin Stadium and (b) plan view of the upper deck of the U.W. Stadium.

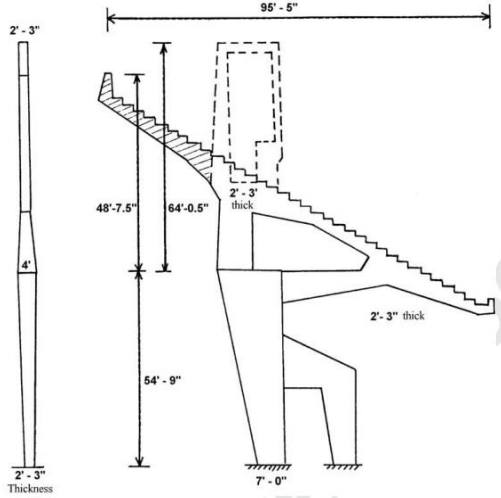


Fig. 2. Typical reinforced concrete frames supporting upper deck of U.W. Stadium.

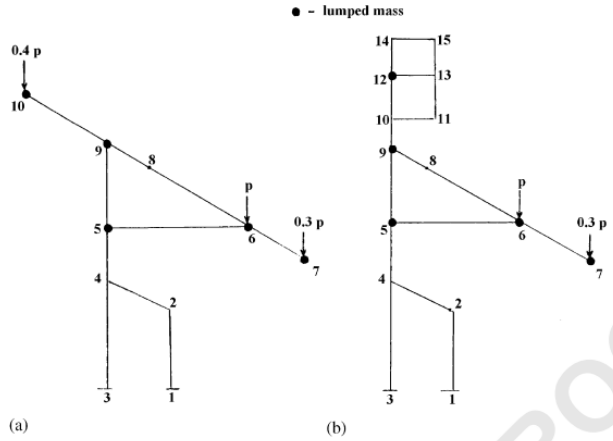


Fig. 3. Finite element models of the supporting frames.

A modal analysis identified the first seven natural frequencies and 3D mode shapes.

The natural frequencies were closely spaced, making it difficult to pinpoint a single dominant mode.

Acceleration data recorded during football games showed two types of responses:

Transient/random vibration from events like touchdowns (Fig. 4a).

Resonant vibration caused by coordinated crowd jumping (Fig. 4b).

Spectral analysis (Fig. 5) showed that resonant response was dominated by a single mode at about 2.3 Hz.

The engineers concluded that the upper deck can be accurately approximated as a single-degree-of-freedom system for resonance calculations.

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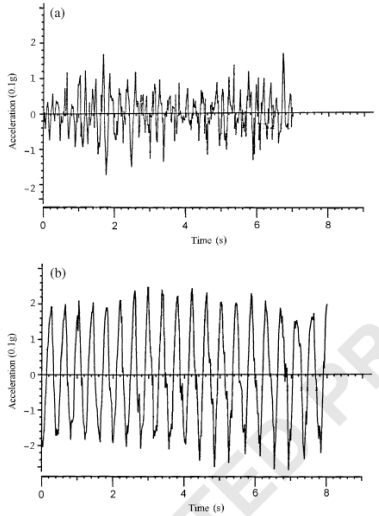

Fig. 4. (a) Acceleration time–history (touchdown response) and (b) acceleration time–history (harmonic vibration).

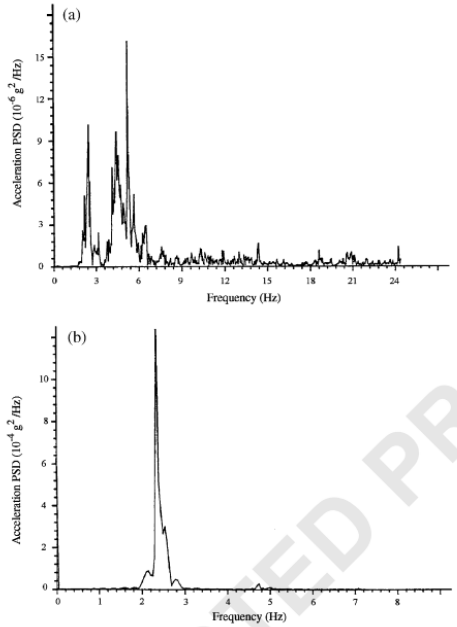

Fig. 5. (a) Response spectrum (touchdown response spectrum) and (b) Response spectrum (resonance spectrum).

Upper deck vibration closely matched a sinusoidal response during coordinated jumping. When spectators stopped, the motion showed typical exponential decay, consistent with lightly damped structures.

Damping ratio was estimated at 2–3% of critical from the decay of peak amplitudes. The modal natural frequency was measured to be approximately 2.3 Hz, which aligns with the frequency range of rhythmic crowd jumping and makes resonance likely.

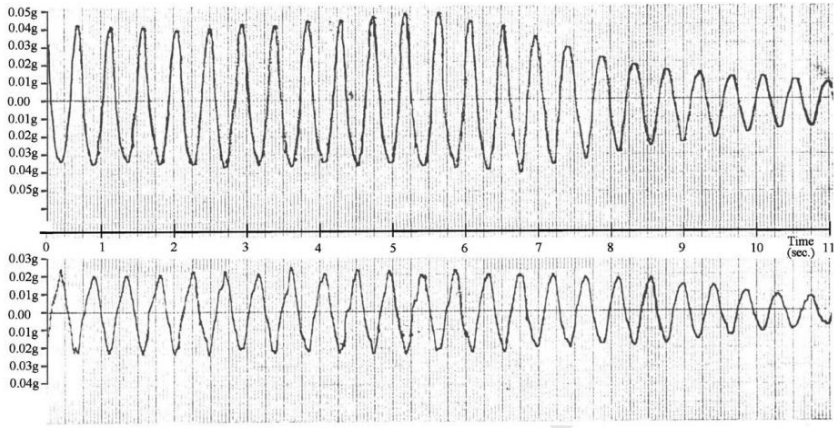

Fig. 6. A typical vibration decay curve ($\xi = 2.3\%$, $f = 2.3$ Hz).

Conclusion

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How Engineers Determined the Tradition Was Safe

- Field measurements showed that upper-deck accelerations during “Jump Around” were well below structural safety limits for concrete and steel stadium systems.
- The dominant vibration at about 2.3 Hz matched a structural mode but produced controlled, predictable motion with low damping (about 2-3% of critical), typical for large seating decks.
- The structure behaved like a single-degree-of-freedom system, making its dynamic response easy to model and accurately assess.
- No signs of fatigue, cracking, or progressive structural degradation were found in follow-up inspections.
- Engineers concluded that the vibrations, while noticeable, were non-damaging, giving strong evidence the tradition posed no structural danger.

How the Tradition Continued After the Study

- After the 2003 evaluation confirmed safety, the university formally reinstated "Jump Around" during home games.
- The tradition has since become a signature event of Camp Randall Stadium and a major part of Wisconsin game-day culture.
- Stadium renovations (2005, 2011, 2022) included updated structural analyses, each confirming adequate capacity for dynamic crowd loads.
- The event remains monitored periodically, but no engineering concerns have emerged in the two decades since the original study.

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Thank You!!
Questions?